CHAPTER

6

Case Selection: Treatment Planning

The primary goal of operative dentistry is to provide functional stomatognathic system along with pleasing esthetics. Such a goal can only be achieved with sound foundation, which in any treatment regime is 'treatment planning'. In operative dentistry, the treatment planning is restricted to management of dental caries, surrounding periodontium and improving esthetics of the individual. Prior to any treatment plan, the operator must arrive at proper diagnosis and the probable prognosis of the disease; the same should be made clear to the patient. Efficiency, profitability and also patient's satisfaction are the results of well-planned procedures. The treatment results are a byproduct of the clinicians' knowledge, planning and execution of the procedures. It is essential to consider patient's needs, attitude and willingness to undergo the specific treatment regime. The patient's general health and the medical conditions should also be taken into account during planning a case for operative procedure. Once mutual understanding is reached, treatment can be executed, instructing the patient regarding the maintenance phase.

CASE SELECTION

Case selection implies thorough conversation with the patient prior to any operative procedure. It includes personal, medical and dental history. Conversation is an art. We must learn how to converse and to extract patient feelings. One must be a patient listener. First, do not ask any question. Let the patient feel comfortable and reveal his/her problems in detail. Second, during conversation, the operator should be very attentive; he/she should not indulge in cross-talk with the assistant or the nurse. Let there be one-to-one conversation. Once the patient feels comfortable, other questions can be asked.

Personal History

Personal and family history includes patient's habits, awareness towards general and oral health. Attitude of parents towards oral care is also important. Certain conditions, which are likely to be inherited, should be explored. The socioeconomic status of the family should also be noted.

Medical History

The detailed information regarding systemic diseases and treatment taken, history of X-radiations and allergies is important prior to start of any restorative treatment. Physiological changes associated with aging also influence treatment regime. Intake of certain medicines, which reduce salivary flow and/or alter the basic defense mechanisms of the body, any allergy to drugs, past history of drugs taken should be noted.

Dental History

Dental history provides information about the patient's awareness towards oral health. Any allergy to anesthetics and local antiseptics should also be noted. Past treatment of any tooth and its complication, if any, has an important impact on the treatment plan. The frequency of patient's visit to dental clinic is also noted.

FORMULATING TREATMENT PLAN

Formulating treatment plan implies providing masticatory system that is functionally adequate and esthetically pleasing. The appropriate plan maintains the balance between patients expectations and the ethical commitment of the operator. When in doubt, or patient cannot be reconciled, choose the ethical plan for the long-term benefit of the patient. The patient has to be convinced that the plan selected will definitely resolve his/her problem with long-term success. The probable barriers, such as cost, time, etc. should be managed accordingly [informed consent should be obtained from the patient (guardian)]. No treatment plan is considered 'concrete' and may evolve over time to suit patients' expectations and the changed circumstances.

Table 6.1: Overview of the treatment protocol plan

Formulating treatment plan: Key features
• Minimizing stress for the patient and the doctor as well • Logical discussion of various treatment options • Potential to achieve patient's expectations • Minimizing risk of legal problems • Plan to manage complications, if any (contingency plan) • Effective use of patient's time in clinic (no inconvenience to patient) • Expected prognosis of the treatment plan • Minimizing risk of revenue loss to the clinic • Ethical contribution to planning and also execution of treatment plan
Case selection: Key features
• Collect all the relevant information by means of case history and clinical examination • Identify list of problems • Carry out consultation with patient's physician; prepare clinic for providing extra care • Consider treatment option, priorities and likely prognosis • Formulate treatment plan in consultation with patient • Prioritize emergency phase, disease control phase and maintenance phase • Maintain 'Can I help you' attitude before, during and after the treatment • Geriatric and/or compromised patients and patients with alarming systemic diseases be treated with sympathy.

Table 6.1 displays overview of the treatment protocol plan during restorative procedures.

The treatment planning is an essential component of successful restorative outcome. It can be finalized in one visit or may take more time to finalize the plan.

Two aspects, the financial management and the time management are important while planning treatment. Inclusion of other specialities (orthodontics, periodontics, endodontics) should also be considered, if need be. The treatment plan must include comprehensive assessment (both subjective history taking and objective clinical examination and diagnostic tests) and alternative option depending upon patient's priorities. The associated complication, if any, should be discussed in detail and the informed consent be obtained from the patient/guardian.

The management of treatment plan is summarized in Table 6.2. The treatment sequence (stages) may be planned as follows:

- Symptomatic relief of pain
- Emergency, if any (verify pulp involvement)
- Clinical examination of soft and hard tissues
- Assessment of caries risk
- Assessment of fractures/tooth substance loss
- Selection of appropriate restorative modality
- Restoration of the tooth with leakage-resistant material
- Measures to prevent the recurrence

Flowchart 6.1 displays the sequence of treatment planning.

Table 6.3 summarizes preferred treatment options and material for different tooth conditions.

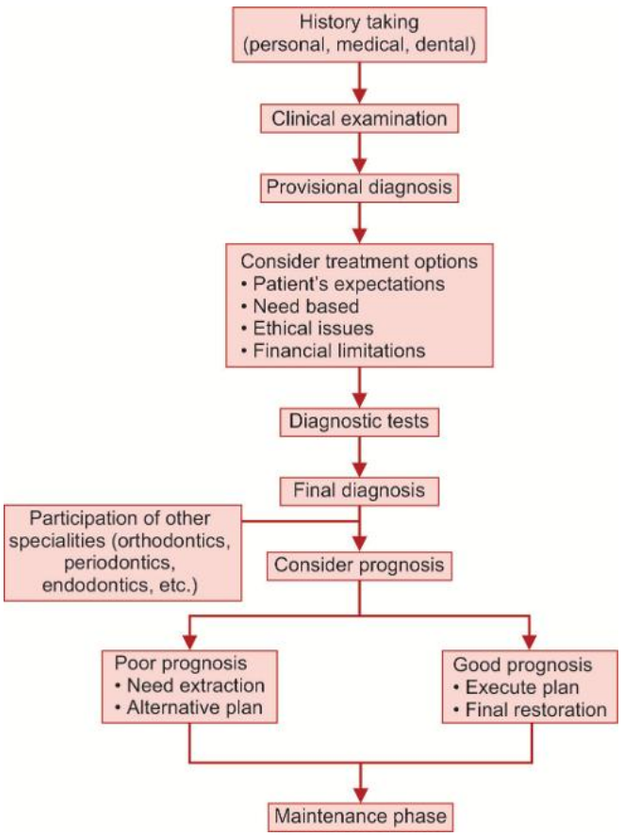
Table 6.2: Management of treatment plan

Treatment plan	*Management*
Emergency	Control of acute pain, gingival bleeding, etc. Pulp treatment; Physician's consultation, if required.
Restorative (normal)	Selection of material and treatment modality
Restorative (complex)	Indirect restorative modalities
Occlusal analysis	Assess response to operative procedures. Evaluate clinically and also study casts
Preventive	Oral hygiene instruction, fluoride application, chlorhexidine mouthrinses, etc.
Maintenance	Review oral hygiene, periodontal health, caries activity, occlusion, etc.

Table 6.3: Preferred treatment options and material for different tooth conditions

Tooth condition	*Preferred treatment options*	*Preferred material*
White spot (opacity)	Remineralizing pastes, oral hygiene instruction, periodic evaluation	Remineralizing agents
Incipient caries	Resin infiltration, sealant application	Low monomer composites/glass ionomer cements
Caries leading to cavitation	Direct or indirect restoration depending upon the destruction	Amalgam, cast metal, composites
Defective restoration	Repair or replacement depending upon the defect	Repair with same material; replace with other alternatives
Tooth fracture	Restore or root canal treatment depending on severity of fracture	Composites
Post-endodontic restoration	Onlay or crown	Cast metal, composite, ceramic

Flowchart 6.1: Sequence of treatment plan



CLINICAL EXAMINATION

After initial conversation, the patient is made to feel comfortable on the dental chair. In a conducive environment, the patient is explored of his/her chief complaints; followed by examination of soft and hard tissues.

Chief Complaint

The patient should be encouraged to discuss all aspects of his/her ailment. The common complaint is 'pain'; it may be esthetic problems due to discolored/malaligned/fractured teeth, etc. Immediate alleviation of pain is mandatory, whenever there is a complaint of pain. The cause of pain, its onset, duration, character and the related factors should be assessed, initiating the required treatment.

Examination of the Soft Tissues

A routine pattern of visual examination is followed palpating the cheeks, vestibule, mucosa, lips, lingual and facial alveolar mucosa to the palate, tonsillar area, tongue and floor of the mouth.

The examination of the gingival color, form, level of attachment and depth of gingival sulcus may reveal variations from the normal. These variations in an otherwise normal oral cavity are indicative of systemic diseases. Certain local conditions, like improper restorations may aid in gingival problems. Before undergoing any operative procedure, gingival health should be restored. Healthy and inflamed gingiva are shown in Fig. 6.1A, B.

Examination of the Hard Tissues (Teeth)

The examination of teeth includes:

i. Number, form and morphology
ii. Color
iii. Restorations
iv. Proximal relationships (contact and contour)
v. Attrition, abrasion and erosion
vi. Occlusion
vii. Factures
viii. Vitaliy of tooth
ix. Carious lesions

i. ***Number, form and morphology:*** The number of teeth is more relevant during young age, especially during mixed dentition period. Congenitally missing and supernumerary teeth (Fig. 6.2A, B) are noted. The total number of teeth present at any particular age is also indicative of the patient's interest in dental care and accordingly aids in devising a suitable

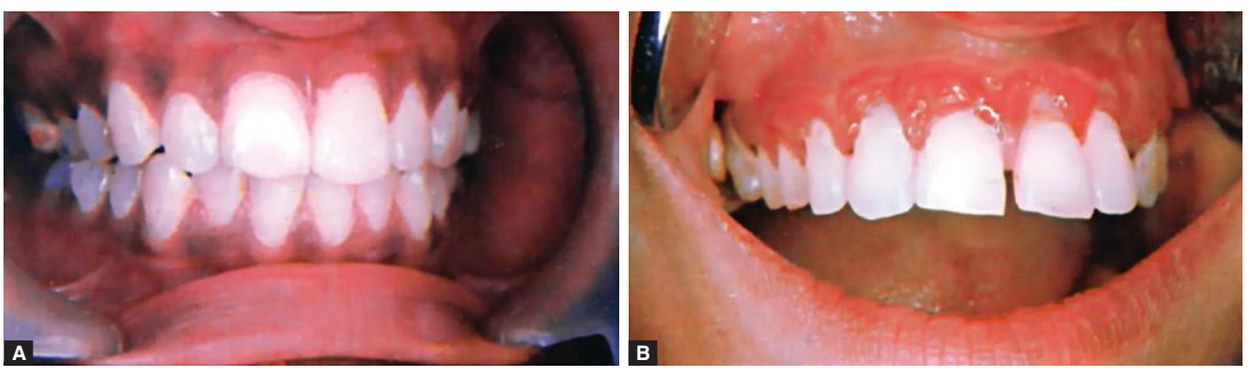


Fig. 6.1: Evaluation of gingiva: **(A)** Healthy gingiva; **(B)** Inflamed gingiva

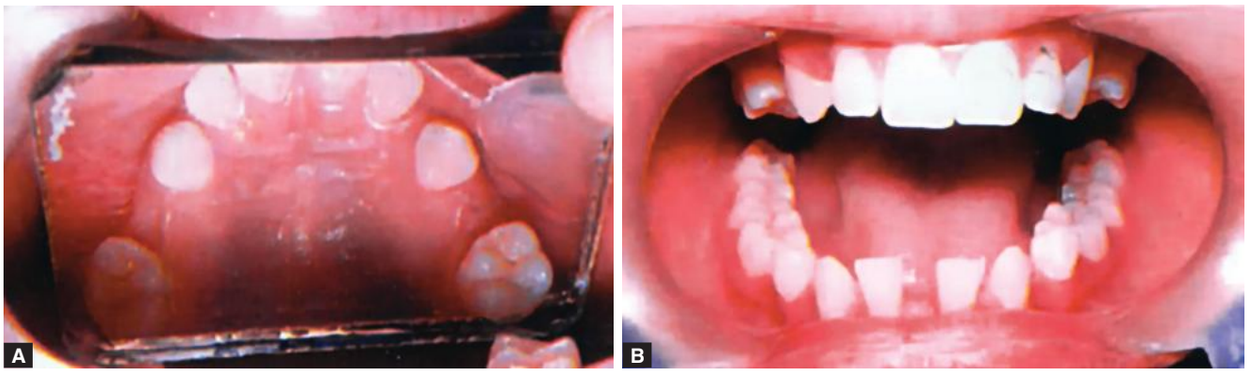


Fig. 6.2: Congenitally missing teeth: **(A)** Maxillary premolars; **(B)** Mandibular lateral incisors

treatment plan. Teeth can show microdontia or macrodontia, gemination, dens in dente, amelogenesis imperfecta, dentinogenesis imperfecta, etc. Morphology of the teeth, many a times, is indicative of a systemic disease; e.g. mulberry molars in syphilis. The operator, may modify the 'square' type molar to its proper shape so as to avoid periodontal problems. Certain morphological changes also occur in hypoplastic teeth, which are of great importance during restorative procedures.

ii. *Color:* Abnormal tooth color is indicative of certain systemic diseases. Non-vital teeth exhibit altered color. Teeth with systemic diseases, like erythroblastosis fetalis and congenital porphyria also show abnormal color. Other conditions exhibit color change are amelogenesis imperfecta, dentinogenesis imperfecta, fluorosis and intake of tetracycline during pregnancy and infancy (Fig. 6.3A, B). If any localized change is visible in enamel color (say opacity), it may be indicative of initiation of caries process or hypoplastic area.

iii. *Restorations:* The following features should be evaluated during examining the oral cavity for restorations:

- Number of restored teeth
- Restorative material used
- Extent of the restoration
- Fractured restoration(s)
- Marginal ditching
- Secondary caries
- Stained margins
- Discolored restorations
- Overhanging restoration
- Voids
- Marginal ridge incompatibility

iv. *Proximal relationships (contacts and contours):* Maintenance of proper contact between two teeth and their exact location is of utmost importance in restorative dentistry so as to protect and stimulate the underlying interdental gingiva. Loose contacts lead to food impaction, thereby leading to periodontal problems (Fig. 6.4A). Conversely, very tight contact deteriorates the periodontal ligament of the adjacent teeth.



Fig. 6.3: Tooth discoloration: **(A)** Fluorosis; **(B)** Tetracycline stains

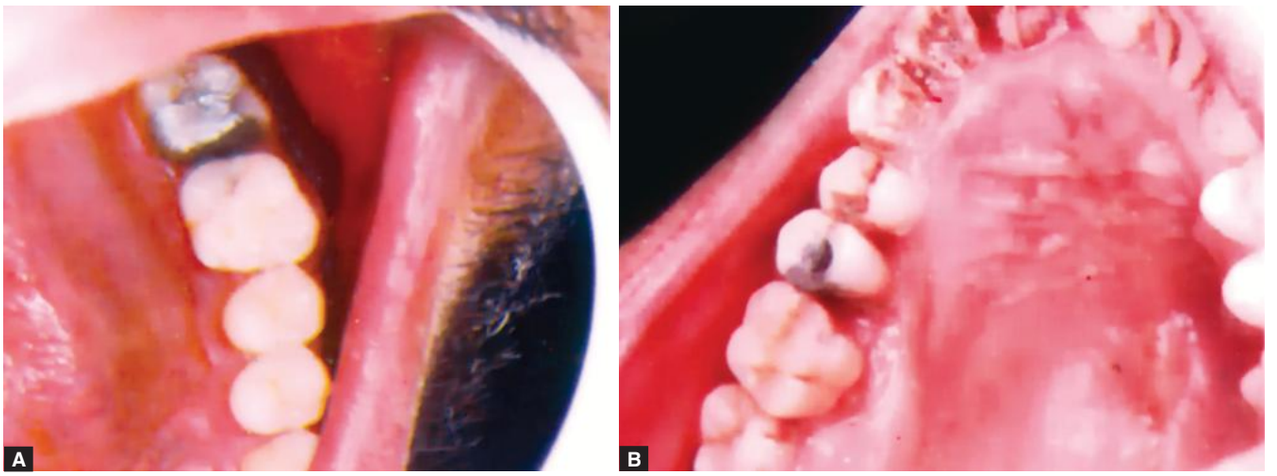


Fig. 6.4: Evaluation of contacts and contours: **(A)** Open contact; **(B)** Improper embrasure

Restoration of proper contour is one of the major concerns in operative dentistry. Over-contoured and undercontoured restorations are harmful to the underlying periodontium. Improper facial and lingual contours invite plaque accumulation leading to thickening of gingival margins and other periodontal problems.

During restoration of proximal lesions, care should be taken to have the proper form of embrasures (Fig. 6.4B).

v. ***Attrition, abrasion and erosion:*** Attrition is the wear of enamel due to masticatory forces, which may be physiologic due to aging or pathologic due to bruxism. Abrasion is the loss of tooth surfaces usually associated with mechanical means when continued for longer duration. Erosion is the loss of enamel and may be dentin, subsequent to acidic dissolution or can be idiopathic. The etiology of erosion should be looked for, since continuous erosion leads to failure of restoration.

vi. ***Occlusion:*** Improper occlusion leads to many problems, like periodontal problems, muscle tenderness, temporomandibular joint dysfunctions, etc.

The deviation from normal occlusion has its impact in planning the type of restoration. Premature contacts and excessive stresses during lateral movements of mandible should be noted before planning for the future restorations.

vii. ***Fractures:*** Fractured teeth (partial/complete), should be noted during clinical examination (Fig. 6.5). Complete fracture of only crown or along with the roots is evident with most of the large mesio-occluso-distal restorations.



Fig. 6.5: Fractured maxillary lateral incisor

Occasionally, only fracture lines are present which may or may not extend up to the underlying pulp. This is known as 'cracked tooth syndrome'. Patients can present with symptoms such as sensitivity to release of biting pressure, thermal sensitivity or even lancinating pain, which cannot be correlated with the local findings and the radiographs.

viii. ***Vitality of tooth:*** The restorative treatment plan depends upon the vitality of tooth (pulp vitality). Pulp vitality tests are important tools in our diagnostic armamentarium, since these tests not only determine the vitality of the tooth, but also the pathological status of the pulp. (For details of pulp vitality tests, the readers are advised to refer 'Essentials of Endodontic' 2nd Edition by Vimal K Sikri.)

ix. ***Carious lesions:*** The detection of incipient carious lesions and the caries activity processes prior to tooth destruction are mandatory for an effective diagnosis and treatment planning.

No single test has been developed which can be considered as 'absolute' for caries diagnosis. The concept of prediction of risk of caries prior to its

Table 6.4: Case Selection: Oral health assessment tool

Oral tissues/features	*0 = Healthy (normal)*	*1 = Minor changes*	*2 = Major changes*
Lips and gums	Smooth, pink, moist No bleeding gums	Dry, chapped, inflamed at corners: slight ulceration of gums	Swollen, ulcerated patch; bleeding or ulcerated at corners
Tongue	Normal, moist, pink	Patchy, fissured, selective coatings	White, ulcerated, or swollen
Saliva	Watery and free-flowing	Dry, little saliva present	Tissues parched, little or no saliva present, thick in consistency
Teeth	No decay or fracture	A few decayed or broken teeth; worn teeth	Many decayed or broken teeth, severe attrition
Oral hygiene	Clean oral cavity	Food particles/tartar Present with gingivitis	Food particles/tartar with periodontitis; severe halitosis
Dental pain	No pain	Minor discomfort	Patient complain of specific pain

The operator, after initial examination, categorize patients after scoring for each tissue/feature examined (0: Normal; 1: Minor changes; 2: Major changes). Total score will be out of 12 (maximum score 2 for six tissues/features). Scoring for individual tissue/feature is preferred for treatment planning, rather than the cumulative score.

occurrence has gained importance. Individuals are identified according to their 'caries risk' levels. Certain tests are undertaken which foretell the person's susceptibility or risk to caries, known as caries activity tests.

An oral health assessment tool is devised to record oral findings (Table 6.4).

A chart is also prepared for record purpose (Fig. 6.6).

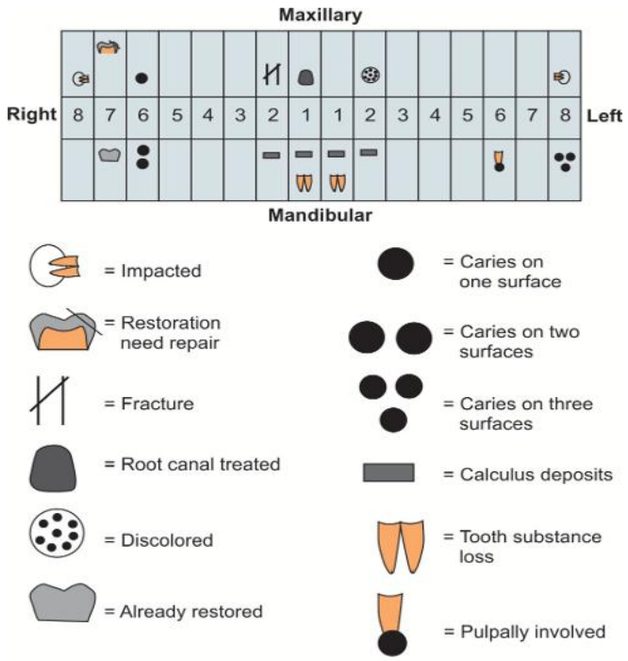


Fig. 6.6: Chart evaluating patients' oral health

CARIES RISK ASSESSMENT

Caries risk assessment is the procedure to predict the future caries development before the classical onset of disease.

'Caries susceptibility' refers to the inherent tendency of the host and the target tissue (the tooth) to be affected by caries.

Factors affecting evaluation of caries risk:

- Morphology and chemistry of teeth (arrangement and number of teeth, occlusal relations, retentive areas, enamel maturation, etc.).
- Biological and chemical factors (localization and composition of plaque, microbial activity, etc.).
- Salivary factors (buffering capacity, immunological factors, oral clearance, etc.).
- Diet (frequency, amount and quality of carbohydrates taken, intake of fluorides, etc.).
- Epidemiology (family caries experience, caries experience in deciduous teeth).
- Systemic conditions (diseases that may influence oral health).

Risk assessment is carried out by:

- Dental saliva pH indicator
- Plaque pH check kit
- Saliva buffer check kit
- Clinpro caries diagnosis L-pop
- Caries risk assessment tool
- Traffic light matrix model

Clinpro caries diagnosis L-pop measures the lactic acid production and evaluates the microbial flora of the mouth by sampling the microflora of the tongue.

American Academy of Pediatric Dentistry (AAPD) classifies risk in infants, children and adolescents [AAPD Caries risk assessment tool (Table 6.5)]. This tool was based on following factors:

Table 6.5: AAPD caries risk assessment tool

Caries-risk indicators	*Low risk*	*Moderate risk*	*High risk*
Clinical conditions	• No carious teeth in past 24 months • No enamel demineralization • No visible plaque; no gingivitis	• Carious teeth in past 24 months • Area of enamel demineralization • Gingivitis	• Carious teeth in past 12 months • More than 1 area of enamel demineralization (white-spot lesion) • Visible plaque on anterior teeth • Radiographic caries • High titers of mutans streptococci • Wearing dental or orthodontic appliances • Enamel hypoplasia
Environmental characteristics	• Optimal systemic and topical fluoride exposure • Consumption of sugar or foods associated with caries initiation, primarily at meal times. • High socioeconomic status • Regular use of dental care	• Suboptimal systemic fluoride exposure with optimal topical exposure • Occasional (usually once or twice) between meal exposures to sugar or foods associated with caries • Mid-level socioeconomic status (eligible for school lunch program) • Irregular use of dental services; no periodic contact with dental professionals	• Suboptimal topical fluoride exposure • Frequent (usually thrice or more) between meal exposures to sugar or foods associated with caries • Low-level socioeconomic status • Active caries present in the mother • No usual source of dental care; never visited dental clinic
General health conditions		• Children with special health care needs • Conditions impairing saliva composition/flow	• Children suffering from mental disorders

a. Clinical conditions
b. Environmental characteristics
c. General health conditions

A traffic light matrix (TL-M) model offers a systematic approach to the assessment of all the risk factors, which contribute to caries activity. Information is used to produce a risk profile from which a treatment regime can be developed. The model is based on the fact that the caries process is multifactorial and it will be driven by changes in the oral environment. The concept of using the colors red, yellow and green in a traffic light color system to convey the different levels of risk has been routinely used. The traffic light does not attempt to predict caries incidence, but rather it alerts the clinician to the presence of risk factors that are capable of changing the oral environment. The TL-M model allocates a threshold value for each risk category. If the information elicited from questioning or by clinical testing yields results, which exceed the predetermined threshold values, the model alerts the clinician to a possible problem. The model investigates nineteen risk factors and scores a red, yellow or green light for each risk factor depending upon predetermined criteria. The system uses a specially designed form to record risk factors.

The second element of the TL-M model is matrix (Table 6.6). This is designed as a means of assessing

Table 6.6: Matrix component of traffic light matrix model

	Disease status		
	1	*2*	*3*
Attitude	a		
	b		
	c		

1. No current disease
2. Need for repair or maintenance
3. Active disease

a. Yes
b. May be
c. No

the patient's present disease status and attitude to maintaining his/her own dental health. It is a very useful measure of the patient's ability or willingness to comply with treatment directives. Attitude towards dental health is scored as a, b or c on the vertical axis of the grid and the current disease status is scored as 1, 2 or 3 and is recorded on the horizontal axis.

The 19 risk factors, grouped under five headings used in the TL-M model, are:

i. *Saliva (six criteria)*
   - Ability of minor salivary glands to produce saliva
   - Hydration of resting saliva
   - Consistency of unstimulated saliva
   - pH of unstimulated saliva
   - Stimulated saliva flow rate (viscosity)
   - Buffering capacity of stimulated saliva

ii. *Diet (two criteria)*
   - Number of sugar exposures per day
   - Number of acid exposures per day

iii. *Fluoride exposure (three criteria)*
   - Exposure via water
   - Exposure via toothpastes
   - Exposure via professional treatment

iv. *Oral biofilm: Plaque (three criteria)*
   - Composition (mutans count)
   - Activity of bacteria
   - pH

v. *Modifying factors (five criteria)*
   - Past and current dental status
   - Past and current medical status
   - Compliance
   - Lifestyle
   - Socioeconomic status

CARIES ACTIVITY TESTS

Caries activity tests are primarily undertaken to evaluate the susceptibility of the individual towards caries. 'Caries activity' refers to the increment of active lesions (new and recurrent) over a period of time. Caries activity is the measure of the speed of progression of caries lesion. Caries activity tests measure the degree to which the local environmental challenge (dietary and microbial factors) favours the probability of initiation of caries.

The tests are based on the hypothesis that oral micro-organisms and fermentable carbohydrates are causative agents in dental caries and the retention of carbohydrates in oral cavity favour growth of micro-organisms.

Caries activity tests help

- Identify high-risk individual/groups
- Determine the need of preventive regimes for particular individual/groups.
- Ensuring low level of caries activity before extensive restorations
- Monitor the effectiveness of oral health education programs

Objectives

For clinicians

- To determine the need for caries control measures.
- To determine optimal time for expertise restoration.
- Aid in determining the results of the preventive measures undertaken.
- Orthodontists can be cautioned during the treatment.

For research workers

- Aid in selection of cases for the study of caries.
- Helps in screening of patented therapeutic agents.
- Serves as an indicator of period of exacerbation.

Ideal requirement of caries activity tests

- Should have sound theoretical basis
- Should be accurate as regards duplication of results
- Should be simple and inexpensive
- Should be less time consuming

The routinely used tests are as follows:

A. Lactobacillus Colony Count Test

This test is of a historical interest; limited mainly to research. It quantitatively evaluates lactobacilli bacteria in the saliva.

Procedure

The patient is asked to chew paraffin before breakfast and the saliva is collected in a bottle. A 1:10 dilution is prepared by pipetting 1.0 mL of this saliva sample into 9.0 mL tube of sterile saline solution. This is shaken and 1:100 dilution is made by pipetting 1.0 mL of 1:10 dilution into another 9.0 mL tube of sterile salt solution.

Then 0.4 mL of each dilution solution is spread on the surface of an agar plate. The plates are labeled and incubated at 37°C for 3–4 days. Quebac counter is used to count the number of colonies.

Advantages

- Simple
- Useful for screening larger groups

Disadvantages

- Inaccurate for predicting onset of caries
- Does not exclude the possibility of growth of other aciduric micro-organisms
- Complex equipment required
- Growth of colonies takes longer time
- Counting of colonies is tedious
- Results take several days
- Costly

Table 6.7: Number of lactobacilli and caries activity

Number of lactobacilli		*Caries activity*
0–1000	+	Little or no caries
1000–5000	+	Slight caries
5000–10000	+ +	Moderate caries
10, 000 and more	+ + +	Marked caries

It is observed that the colony count up to 1000 is unimportant; between 1000 and 5,000 is indicative of slight caries activity; between 5,000 and 10,000 is moderate caries activity and count over 10,000 is associated with high caries activity (Table 6.7).

B. Snyder's Test

Snyder introduced a relatively simple colorimetric test for the estimation of relative number of lactobacilli in the saliva. The test has been widely used as a diagnostic tool as well as to evaluate the patient's acceptance of dietary changes or preventive measures designed to reduce dental caries.

Procedure

Saliva is collected as is given in Lactobacillus test. A tube of Snyder glucose is melted and then cooled to 50°C. 0.2 cc of collected saliva is pipetted into 10 mL of medium and is incubated at 37°C for 72 hours. Rate of color change is indicative of the caries activity.

Glucose agar medium with bromocresol green
+
0.2 mL of saliva
↓ Incubated at 37°C

Color changes in		
24 hrs	48 hrs	72 hrs
Yellow	Yellow	No color change
Marked caries susceptibility	Definite caries activity	Caries inactive

Advantages

- Test is easier and an acceptable method of educating people.
- Color changes can be noted easily as compared to the counting of bacterial colonies.
- Cost effective

Disadvantages

- Cannot predict the extent of expectancy of caries with any reliability for one individual.
- Color change might not be clear
- Measures acidogenic potential; salivary organisms may not be representative of those in plaque
- Time consuming

C. Salivary Reductase Test

The test measures the activity of the enzyme reductase in salivary bacteria. The saliva is collected as mentioned earlier and mixed with a dye—Diazoresorcinol. The change of color is noted in 15 minutes.

Color Changes

Blue : Non-conducive
Orchid : Slight caries activity
Red : Moderate caries acitivity (high caries activity, if color changes to red prior to 15 miutes)
Colorless : Highly conducive

Advantages

- No incubation required
- Time saving (quick results)

Disadvantage

- Results vary with time (after food intake and after brushing)

D. Alban Test

The Alban test eliminates the need for melting the medium as the medium used is soft and allows easy penetration of saliva. The procedure is convenient as the patient expectorates directly into the tubes that contain the medium.

Procedure

60 gm of Snyder test agar is placed in one litre of water and boiled over low or medium heat. When adequately melted, the molten agar is distributed in tubes, each tube containing 5.0 mL of the test medium. The tubes should be allowed to cool in the refrigerator.

The patient is asked to expectorate directly into the tube, such that sufficient amount of saliva is present to cover the surface of the test medium. The tubes are labeled and incubated at 37°C for 4 days. Observations are made for the color change (bluish green, pH 5.0 to definite yellow, pH 4 or less) and the depth of color change. The volume of saliva, the time of day when saliva is collected and the proximity of this time to the time of eating does not significantly affect the results.

The scores are designated as follows:

- No color change –
- Color change beginning at the top of medium +
- One-half color change from top to bottom + +
- Three-fourths color change from top to bottom + + +
- Total color change to yellow + + + +

The final recording after 72 or 96 hours of incubation is carried out as follows:

- Readings negative for the entire incubation period are labeled 'negative'.
- Readings are labeled 'positive' irrespective of the degree of positivity.
- Slower change or less color change compared to the previous test is labeled as 'improved'.
- Faster change or more pronounced change compared to the previous test is labeled as 'worse'.
- When readings are identical compared to the previous test, it is labeled as 'no change'.

Advantages

- Simple and inexpensive
- Ideal for motivation and education

Disadvantages

- More armamentarium required
- Color change evaluation might be subjective

E. Fosdick's Calcium Dissolution Test

The acid formed when patient's saliva is mixed with powdered enamel incubated at 37°C is used for four hours to evaluate the amount in milligrams of powdered enamel dissolution.

Procedure

Saliva is stimulated by chewing gum (paraffin is avoided because glucose is needed for the test; in case paraffin can't be avoided, additional glucose is added). Part of saliva so collected is analyzed for calcium content. The rest is kept in a tube containing 0.1 gm of powdered enamel for four hours at body temperature. After that, it is analyzed for calcium content. The caries activity is directly proportional to enamel dissolution.

Dewar, later modified the test measuring pH of the saliva–glucose–enamel mixture. (Instead of calcium dissolution, only pH is measured.)

Disadvantages

- Complex test
- Accurate measurement is difficult
- Costly

F. Ora Test

The Ora test evaluates the oral microbial level using sterile milk in the oral cavity. In normal conditions, the bacterial enzyme (aerobic dehydrogenase), transfers electrons or protons to oxygen. Once oxygen gets utilized by the aerobic organisms, methylene blue acts as an electron acceptor and gets reduced to leucomethylene blue. This reflects the metabolic activity of the aerobic organisms.

Advantages

- Not-toxic, simple and inexpensive
- Trained personnel not required
- Can also monitor gingival inflammation
- Good for motivation and education
- Cost-effective

Disadvantages

- Lack of specificity (does not identify a specific group of micro-organisms in a specific disease).
- Does not differentiate between onset and progress of caries.
- Positive signs can be obtained with other infections of oral cavity.

Procedure

10 mL of sterile milk is rinsed in the oral cavity for 30 seconds and the expectorate is collected. 3.1 mL of this milk is transferred to a tube and 0.12 mL of 0.1% methylene blue is added. The tubes are observed every 10 minutes for any color change and also the time of initiation of color change. The higher the micro-organisms, lesser is the time taken for the change in color reflecting higher oral microbial level.

G. Swab Test

Swab test is used in young children. The oral flora is collected by swabbing the buccal surface of the teeth with a cotton applicator. It is then incubated in the medium. The change in pH following 48 hours incubation period is evaluated on pH meter.

Advantages

- No collection of saliva required
- Predict caries increment, particularly in children

H. *Streptococcus mutans* Level Test

The test measures the number of *Streptococcus mutans* colony forming units (CFU)/unit volume of saliva by culturing the plaque samples from discrete sites (occlusal/proximal). Tongue blades/wooden spatulas are used to collect the samples. These are then pressed against *Streptococcus mutans* selective mitis

Advantage

- Frequency of isolation of *Streptococcus mutans* is high prior to initiation of lesions.

Disadvantages

- *Streptococcus mutans* tends to be located as specific sites
- Shelf life of plates is one week; so not convenient for chairside tests.
- Difficult to distinguish between carrier state and cariogenic infection.

salivarius bacitracin agar. The agar plates are incubated at 37°C for 48 hours in CO_2 gas.

The test is interpreted as: Level of *Streptococcus mutans* >100/mL of saliva = Unacceptable

I. Salivary Buffer Capacity Test

Salivary buffer capacity test can be quantitated using either pH meter or color indicator. The test measures the quantity of acid in milliliters required to lower the pH of saliva through an arbitrary pH interval (say from pH 7.0 to 6.0) or the amount of acid/base necessary to bring color indicators to their point.

Procedure

Ten milliliter of stimulated saliva is collected at least one hour after eating. 5.0 mL of this is taken in a beaker. The pH of saliva is adjusted at 7.0 by addition of lactic acid or base. Lactic acid is further added until pH 6 is reached. The amount of lactic acid needed to reduce the pH from 7.0 to 6.0 is a measure of buffer capacity. The number is converted to milliequivalent per liter.

There is an inverse relationship of buffering capacity of saliva and caries. The saliva of individual having sufficient number of carious lesion frequently has lower acid buffering capacity then the saliva of those who are relatively caries free.

Advantage
• Simple and effective

Disadvantage
• May not correlate adequately with caries activity.

J. *Streptococcus mutans* Dip-slide Method

This test comprises Dentocult SM and Cari Screen SM (two different procedures).

The test classifies salivary samples according to estimates of *Streptococcus mutans* colonies growing on modified mitis salivarius agar.

Procedure A (Dentocult SM)

- The patient is asked to chew paraffin wax for five minutes and the resultant stimulated saliva is collected.
- Saliva is poured over agar-coated slide, totally wetting the surface, and excess is allowed to drain off.
- Slides are dried for 10–15 minutes, bacitracin discs are placed in middle of the inoculated agar.
- A CO_2 tablet is inserted in the tube containing the slide, which is then incubated for 48 hours.
- A zone of inhibition (10–20 mm in diameter) is formed around each bacitracin disc.
- *Streptococcus mutans* appears as blue colonies growing within the zone of inhibition.
- The colony density is compared with a model chart.

Procedure B (Cari Screen SM)

- Bacitracin tablet is placed in buffered diluents and allowed to dissolve completely.
- The patient is asked to chew paraffin wax for 15–20 seconds.
- While continuing to chew the wax, the patient expectorates approx. 1.0–2.0 mL of stimulated saliva directly into diluents vial, which is then capped and gently mixed by repeated inversion.
- The dip slide coated on both sides with MSB agar (without bacitracin), is then immersed for a few seconds in the buffered diluent containing bacitracin and patient's saliva.
- A CO_2 tablet is placed in the empty dip slide vial, and two drops of water are added.
- The dip slide is replaced in its vial and cap is securely tightened.
- The vial is placed upright for 48 hours at 37°C in an incubator and then allowed to stand at room temperature overnight.
- The colony density on agar is compared with a reference colony density chart.
- This test correlates with actual laboratory methods of quantifying *Streptococcus mutans* by standard culturing.
- The *Streptococcus mutans* colonies can be recognized by their opaque, highly convex appearance and irregular shape.

K. *Streptococcus mutans* Replicate Technique

This technique localises *Streptococcus mutans* colonies on tooth surface using a solid impression matrix composed primarily of sucrose and commercial gum base.

Procedure

- Imprint of tooth surface to be sampled is obtained by pressing the matrix against it.
- Matrix is washed for several seconds in water to remove non-adherent cells.
- Matrices are placed in liquid broth and incubated at 37°C overnight and examined directly for overgrowth of *Streptococcus mutans* colonies at specific sites.

SYSTEMIC DISEASES INFLUENCING TREATMENT PLANNING

A thorough medical history should be recorded, analyzing the probability of any systemic disease.

Communicable diseases, which may affect treatment planning are to be identified, viz.:

- Herpes
- Hepatitis-B
- HIV
- Tuberculosis
- Mumps
- Oral candidiasis

(Transmissible through body fluids, such as blood and saliva, pose great risk to the doctor and the staff).

Systemic conditions affecting operative treatment are:

- Cardiovascular diseases
- Asthma
- Bleeding disorders
- Diabetes
- Renal diseases
- Human immunodeficiency virus (HIV)
- Pregnancy

The patient's physician should be consulted before initiating any procedure, especially elderly patients and others suspected of any systemic involvement. The systemic conditions, which warrant extra attention before executing treatment planning, are:

a. ***Cardiovascular diseases:*** When a patient has a history of hypertension or taking anticoagulants, such patients are to be treated with utmost care. In any case, where the patient is at risk, the operative procedures should be carried out under antibiotic premedication.

Mouth wash with 5.0% betadine/chlorhexidine is advised before rubber dam placement to avoid any possibility of bacteremia due to injury to gums. Vasoconstrictors should not be administered to patients with unstable angina, recent myocardial infarction, recent stroke, recent arrhythmias and hypertensive patients. Local anesthetic agents without vasoconstrictor or minimal vasoconstrictors are preferred.

Patients with artificial heart valves are considered highly susceptible to bacterial endocarditis. Such patients should be given prophylactic antibiotics before initiating operative procedures.

It is mandatory to check whether hypertension is under control or not, before administering local anesthesia [*even though adrenaline is contraindicated in hypertensives, amount available in 1:100000 solutions is safe. As anesthetics without vasoconstrictors have a less profound effect and shorter duration of action, slow injection (one minute for 1.8 mL) using 26 or 27 gauge needle is preferred*].

b. ***Asthma:*** Asthma, hay fever, skin rashes, etc. indicate hypersensitivity problems and the patients should be managed accordingly. The patient's medication and nebulizer should be kept ready. The rubber dam application should be modified and removed after short interval of time. Clinic set up should be aerosol free.

The patient is thoroughly explored for:

- Whether patient is on medication: If yes, for which problem?
- Whether the medication was prescribed by any qualified doctor?
- Any allergic reactions in the near past.

This helps in identifying patient attitude towards his/her ailments (some medication may be interfering with the antibiotics and analgesics, need to be analyzed).

c. ***Bleeding disorders:*** The bleeding disorders (blood dyscrasias) are not definite contraindications for executing operative procedures; however, the operator should be conscious of the implications of these diseases.

- Check bleeding and clotting time
- Less traumatic procedure should be planned. (*Most challenging is to treat a patient with hemophilia as there is always the risk of puncturing blood vessel and inducing internal bleeding, particularly with inferior alveolar nerve blocks. Care should be exercised in placing rubber dam clamps so that they do not impinge on gingival tissue).*

d. ***Diabetes:*** The controlled diabetic patients, free from complications like renal failure, hypertension and/or coronary atherosclerotic disease, can be treated in routine. The prophylactic antibiotics are prescribed as a standard protocol. The appointments should be scheduled considering the patient's normal meal and insulin schedule.

- Check whether diabetes is under control or not (stage of diabetes).
- Patients with diabetes are prone to infection and delayed healing; hence, necessary precautions should be taken.
- Schedule early morning appointments where the patient has had medication and meals, so that the risk of hypoglycemia is minimized.

e. ***Renal diseases (dialysis):*** Patient with renal failure and other kidney dysfunction may not be able to metabolize certain drugs. Hemodialysis tends to aggravate bleeding tendencies through physical destruction of platelets. It is important to review the patient's undergoing dialysis before initiating operative treatment. The operator should be aware of the fact that hemodialysis removes certain drugs from the circulating blood and can shorten the effect of prescribed medications. Certain drugs may have to be avoided or their doses are to be adjusted. Penicillin V and cephalexin require increase in dosage; however, erythromycin requires no

adjustment. The patient's physician should be consulted as regard specific drug requirements.

f. ***Human immunodeficiency virus (HIV):*** It has been established that the potential of developing hepatitis B/hepatitis C from a needle-stick injury is much higher than developing HIV. A major consideration for the management of patients with HIV/acquired immunodeficiency syndrome (AIDS) is the CD4 lymphocyte count. Patients with CD4 count above 400 mm^3 are considered asymptomatic HIV candidates and can be treated in routine. The operative treatment for HIV infected patients can be initiated in consultation with the medical specialist. The intake of medicine should also be noted and can be modified.

g. ***Pregnancy:*** Unless emergency treatment is warranted, it is advisable to defer elective operative treatment, especially during the first trimester. The second trimester is the safest period during which any procedure can be carried out. Treatment planning should be directed at eliminating pain symptoms during pregnancy. The chair position should be comfortable for the patient.

The postpartum period, wherein the mother breast-feed the child, should also be considered. The clinician should avoid any drug harmful to the infant. Patient's gynecologist should be consulted as regard treatment needs of the patient and also the drug regime. Organogenesis is taking place in first trimester; any medication has to be cleared by the gynecologist. Second trimester is relatively safe. Third trimester is also safe but care has to taken with chair position as the fetus is growing and postural hypotension may develop. To prevent supine hypotensive syndrome in the dental chair, the pregnant woman should have the right hip elevated 10 to 12 cm or placing the patient in a 5.0 to 15% tilt on her left side can relieve pressure on the inferior vena cava. If hypotension is still not relieved, a full left lateral position may be needed.

MAINTENANCE PHASE

After executing the treatment plan (final restoration), the patient is allowed some holding time for re-evaluating the treatment along with time for healing and resolution of inflammation (if any). The patient is kept motivated for maintenance of the treatment opted and also caring for abnormal habits, if recorded earlier.

A regular recall examination is necessary; which provide opportunity to reinforce home care and may need minor adjustments after functional use. Recall is important especially in patients with high caries risk and also with unstable periodontal health.

Finishing/polishing, if required, should be carried out periodically, during the maintenance phase. Many features can be traceable, which might have been overlooked during initial examination. Summarily, the phase maintains the oral health of the patients.

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